

Practice Exam 3

STAT 3113 Regression Analysis

The Boston Housing Dataset

In R, the MASS library contains Boston data set, which has 506 rows and 14 columns, which records medv (median house value) and 13 predictors for 506 neighborhoods around Boston. The data frame contains the following columns:

- crim —per capita crime rate by town.
- zn — proportion of residential land zoned for lots over 25,000 sq.ft.
- indus — proportion of non-retail business acres per town.
- chas — Charles River dummy variable (= 1 if tract bounds river; 0 otherwise).
- nox — nitrogen oxides concentration (parts per 10 million).
- rm — average number of rooms per dwelling.
- age — proportion of owner-occupied units built prior to 1940.
- dis — weighted mean of distances to five Boston employment centres.
- rad — index of accessibility to radial highways.
- tax — full-value property-tax rate per \$10,000.
- ptratio — pupil-teacher ratio by town.
- black — $1000(Bk - 0.63)^2$ where Bk is the proportion of blacks by town.
- lstat — lower status of the population (percent).
- medv — median value of owner-occupied homes in \$1000s. (**dependent variable**)

Load the Data

```
library(MASS)
names(Boston)
```

```
## [1] "crim"    "zn"      "indus"   "chas"    "nox"     "rm"      "age"
## [8] "dis"     "rad"     "tax"     "ptratio" "black"   "lstat"   "medv"
```

Question 1

Conduct a stepwise regression analysis of the Boston data using R to find the “best” predictors of medv. Please use 0.15 for both α -to-remove and α -to-enter. Comment on the output.

```
library(olsrr)
fit_complete<-lm(medv~., data=Boston)
ols_step_both_p(fit_complete, details=FALSE, pent=0.15, prem = 0.15)
```

```
##
##                               Stepwise Selection Summary
```

##	Step	Variable	Added/ Removed	R-Square	Adj. R-Square	C(p)	AIC	RMSE
##	1	lstat	addition	0.544	0.543	362.7530	3288.9750	6.2158
##	2	rm	addition	0.639	0.637	185.6470	3173.5423	5.5403
##	3	ptratio	addition	0.679	0.677	111.6490	3116.0973	5.2294
##	4	dis	addition	0.690	0.688	91.4850	3099.3590	5.1386
##	5	nox	addition	0.708	0.705	59.7540	3071.4386	4.9939
##	6	chas	addition	0.716	0.712	47.1750	3059.9390	4.9326
##	7	black	addition	0.722	0.718	37.0590	3050.4384	4.8818
##	8	zn	addition	0.727	0.722	30.6240	3044.2750	4.8474
##	9	crim	addition	0.729	0.724	28.4180	3042.1546	4.8326
##	10	rad	addition	0.734	0.729	20.2660	3034.0687	4.7895
##	11	tax	addition	0.741	0.735	10.1150	3023.7264	4.7362

Answer:

Question 2

What are the dangers associated with drawing inferences from the stepwise model?

Answer:

Question 3

Use all-possible-regressions-selection to find the “best” predictors of medv. Include the adjusted r-square plot and Cp plot, and illustrate how the choice is made.

```
library(leaps)
all.possible<-regsubsets(medv~., data=Boston, nvmax=13)

bestsubset.summary<-summary(all.possible)
bestsubset.summary$outmat
```

```
##          crim zn  indus chas nox rm  age dis rad tax ptratio black lstat
## 1  ( 1 )  " "  " "  " "  " "  " "  " "  " "  " "  " "  " "  " "
## 2  ( 1 )  " "  " "  " "  " "  " "  "*"  " "  " "  " "  " "  " "
## 3  ( 1 )  " "  " "  " "  " "  " "  "*"  " "  " "  " "  " "  "*"
## 4  ( 1 )  " "  " "  " "  " "  " "  "*"  " "  " "  " "  " "  "*"
## 5  ( 1 )  " "  " "  " "  " "  " "  "*"  "*"  " "  " "  " "  "*"
## 6  ( 1 )  " "  " "  " "  "*"  "*"  "*"  " "  " "  " "  " "  "*"
## 7  ( 1 )  " "  " "  " "  "*"  "*"  "*"  " "  " "  " "  " "  "*"
## 8  ( 1 )  " "  "*"  " "  "*"  "*"  "*"  " "  " "  " "  " "  "*"
## 9  ( 1 )  "*"  " "  " "  "*"  "*"  "*"  " "  " "  "*"  " "  " "
## 10 ( 1 )  "*"  "*"  " "  " "  "*"  "*"  " "  " "  "*"  "*"  " "
## 11 ( 1 )  "*"  "*"  " "  "*"  "*"  "*"  " "  " "  "*"  "*"  " "
## 12 ( 1 )  "*"  "*"  "*"  "*"  "*"  "*"  " "  " "  "*"  "*"  " "
## 13 ( 1 )  "*"  "*"  "*"  "*"  "*"  "*"  "*"  " "  "*"  "*"  " "
```

```
rsq = bestsubset.summary$rsq
adjrsq = bestsubset.summary$adjr2
rss = bestsubset.summary$rss
cp = bestsubset.summary$cp
```

```
p=seq(1, 13, 1)
cbind(p, adjrsq)
```

```
##      p    adjrsq
## [1,]  1 0.5432418
## [2,]  2 0.6371245
## [3,]  3 0.6767036
## [4,]  4 0.6878351
## [5,]  5 0.7051702
## [6,]  6 0.7123567
## [7,]  7 0.7182560
## [8,]  8 0.7222072
## [9,]  9 0.7252743
## [10,] 10 0.7299149
## [11,] 11 0.7348058
## [12,] 12 0.7343282
## [13,] 13 0.7337897
```

```
cbind(p, cp)
```

```
##      p      cp
## [1,]  1 362.75295
## [2,]  2 185.64743
## [3,]  3 111.64889
## [4,]  4  91.48526
## [5,]  5  59.75364
## [6,]  6  47.17537
## [7,]  7  37.05889
## [8,]  8  30.62398
## [9,]  9  25.86592
## [10,] 10  18.20493
## [11,] 11  10.11455
## [12,] 12  12.00275
## [13,] 13  14.00000
```

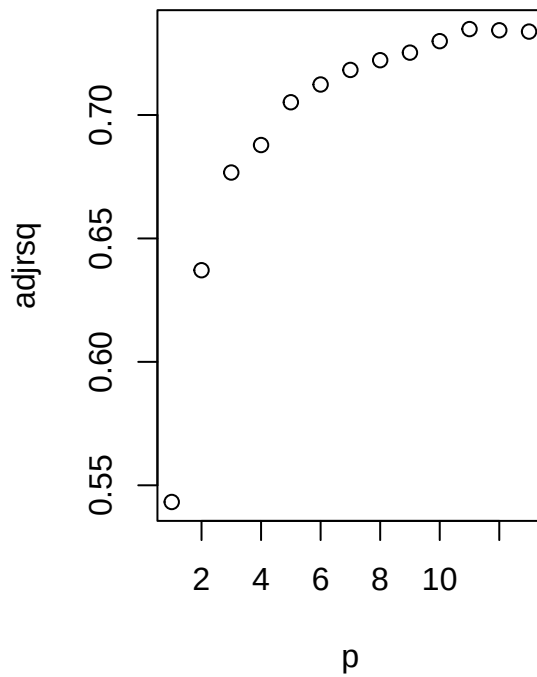
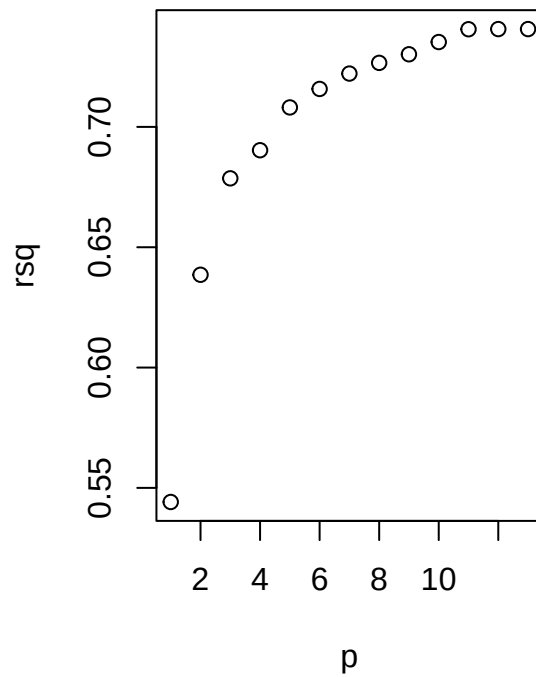
```
cbind(p, rss)
```

```
##      p      rss
## [1,]  1 19472.38
## [2,]  2 15439.31
## [3,]  3 13727.99
## [4,]  4 13228.91
## [5,]  5 12469.34
## [6,]  6 12141.07
## [7,]  7 11868.24
## [8,]  8 11678.30
```

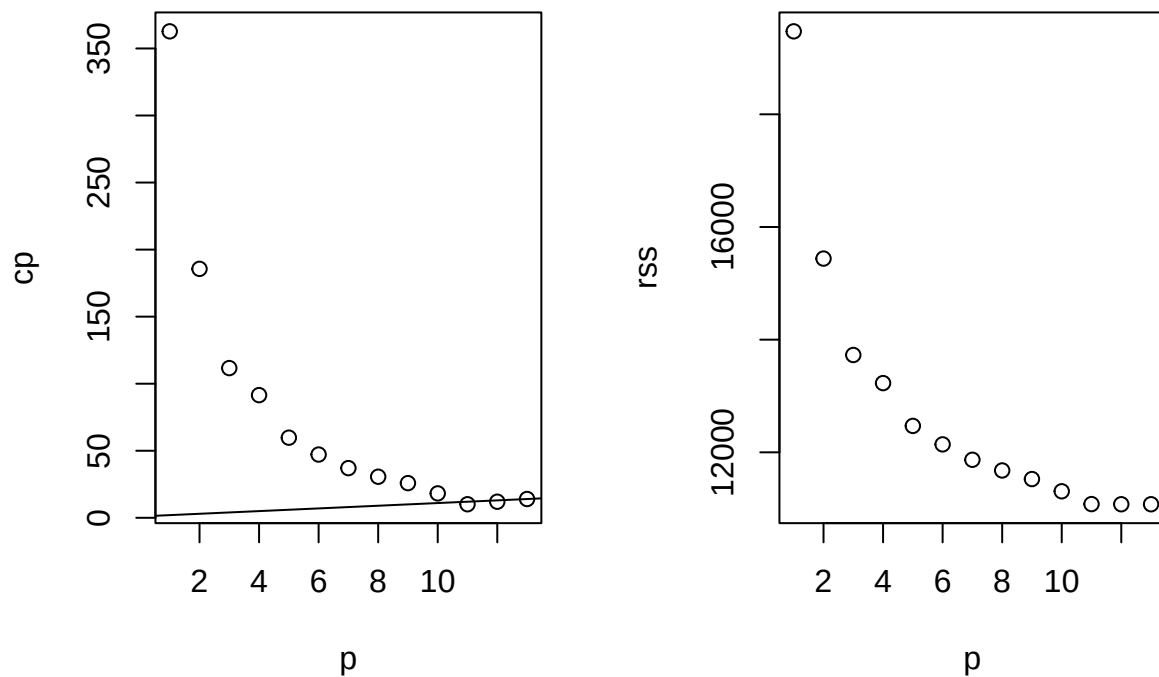
```
## [9,] 9 11526.12
## [10,] 10 11308.58
## [11,] 11 11081.36
## [12,] 12 11078.85
## [13,] 13 11078.78
```

```
### Plot the rsq, adjrsq, rss, cp vs p
```

```
par(mfrow=c(1,2))
plot(p, rsq)
plot(p, adjrsq)
```



```
plot(p, cp)
abline(1,1)
plot(p, rss)
```



```
par(mfrow=c(1,1))
```

Answer:

Question 4

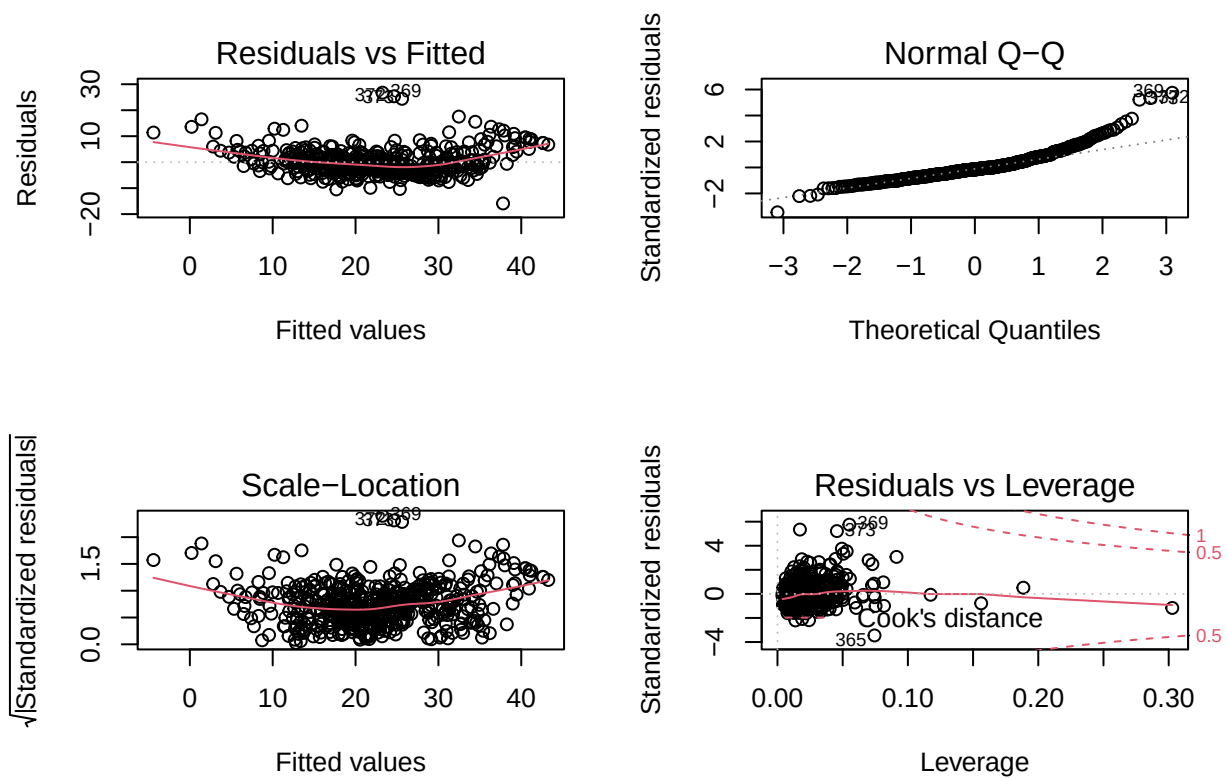
Compare the results in Question 1 and 3, which independent variables consistently are selected as the “best” predictors?

Answer:

Question 5

Fit the first-order linear model with 10 independent variables: crim, chas, nox, rm, dis, rad, tax, ptratio, black, lstat. Plot the residual plots. Comment on the four residual plots one by one on the issues you found. What model adjustments would you recommend?

```
fit = lm(medv~crim + chas + nox + rm + dis + rad + tax + ptratio + black + lstat, data = Boston)
par(mfrow=c(2,2))
plot(fit)
```



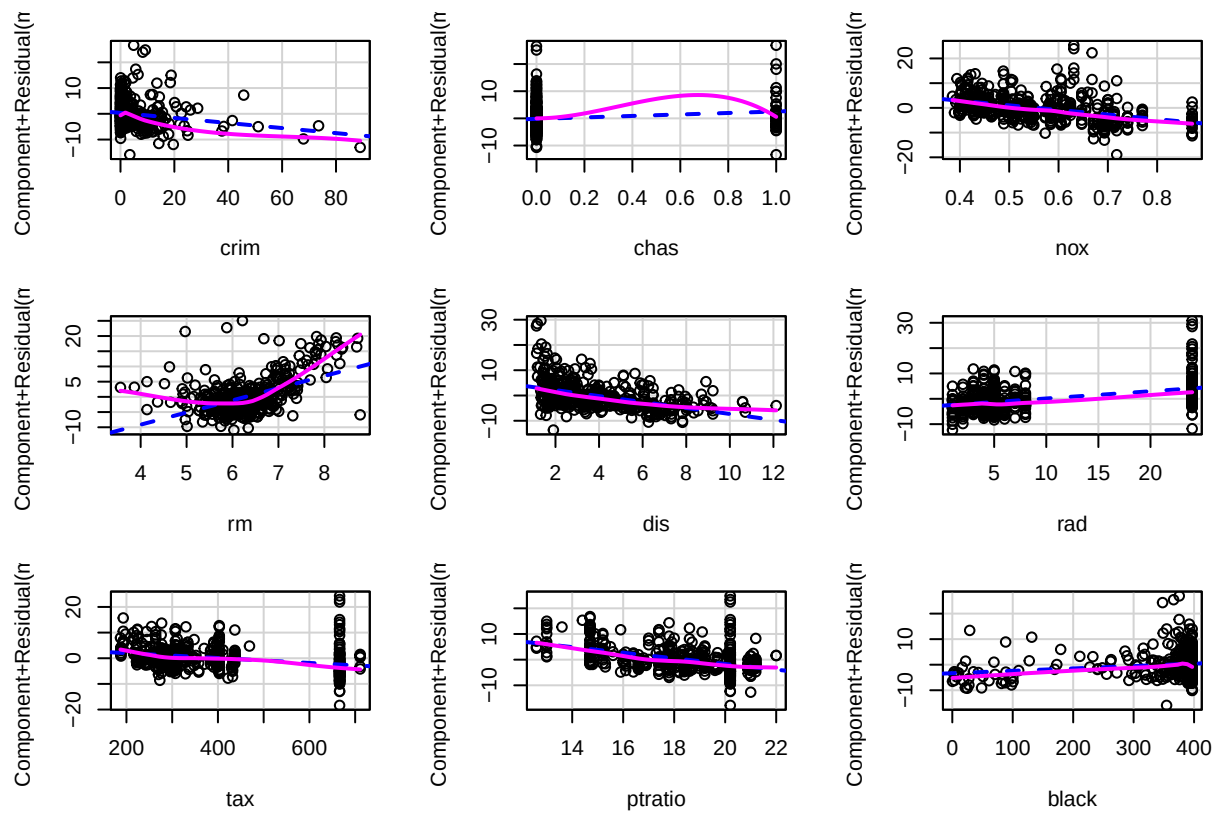
```
par(mfrow=c(1,1))
```

Answer:

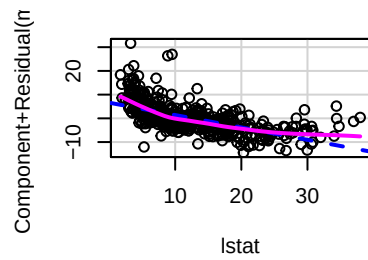
Question 6

Use the following code to plot the partial residual plots. Comment on the partial residual plots. What do the plots reveal the information between medv and the independent variables?

```
library(car)
crPlots(fit)
```



Component + Residual Plots



Answer:

Question 7

Fit the model with 10 variables in Question 5 and add second-order term of lstat, second-order term of rm, and the interaction between rad & lstat, rm & rad, crim & chas, chas & nox.

```
fit.c = lm(medv~crim + chas + nox + rm + dis + rad + tax + ptratio + black + lstat
           + I(lstat^2) + I(rm^2) + rad:lstat + rm:rad + crim:chas + chas:nox,
           data = Boston)
summary(fit.c)
```

```
##
## Call:
## lm(formula = medv ~ crim + chas + nox + rm + dis + rad + tax +
##      ptratio + black + lstat + I(lstat^2) + I(rm^2) + rad:lstat +
##      rm:rad + crim:chas + chas:nox, data = Boston)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.610  -2.026  -0.223   1.604  24.638
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

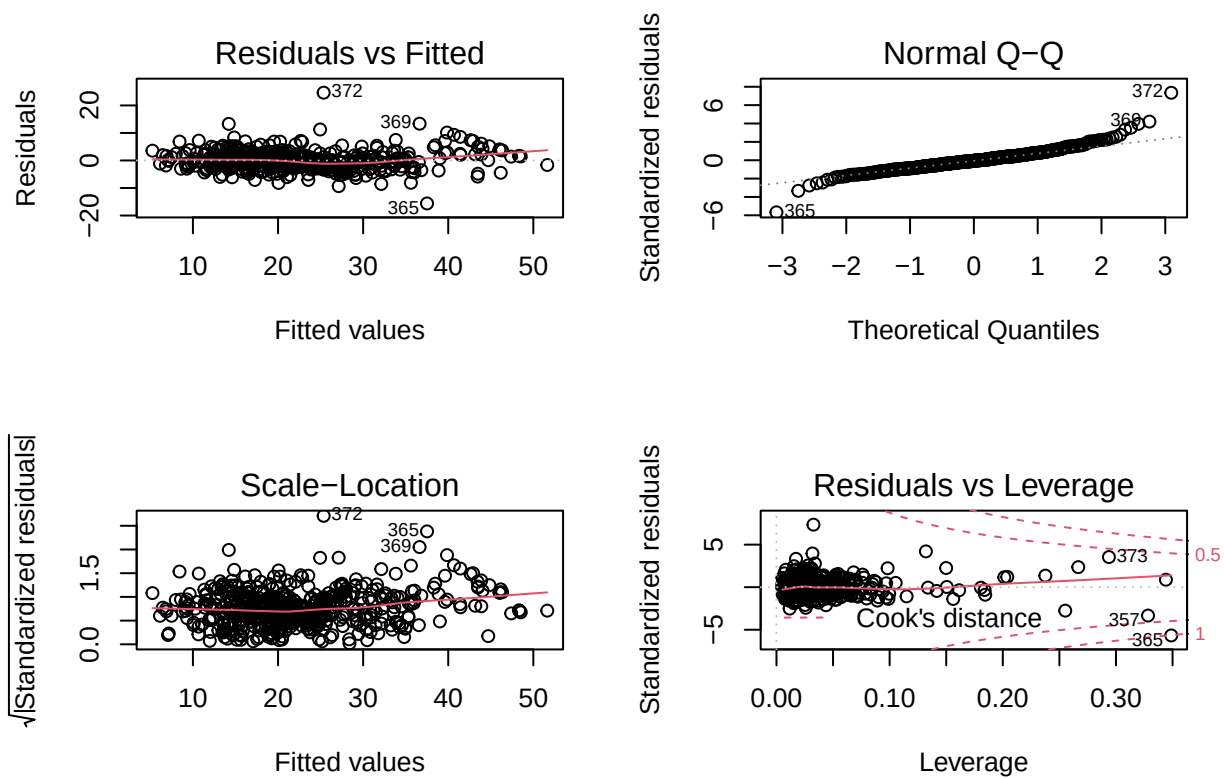


```
## (Intercept) 42.046646 11.485042 3.661 0.000279 ***
## crim -0.111171 0.024327 -4.570 6.19e-06 ***
## chas 13.473038 3.087684 4.363 1.56e-05 ***
## nox -12.363598 2.764480 -4.472 9.63e-06 ***
## rm -4.427683 3.056792 -1.448 0.148126
## dis -0.820218 0.120105 -6.829 2.54e-11 ***
## rad 2.802634 0.264222 10.607 < 2e-16 ***
## tax -0.008726 0.002397 -3.640 0.000301 ***
## ptratio -0.745128 0.090980 -8.190 2.29e-15 ***
## black 0.003524 0.001948 1.809 0.071130 .
## lstat -0.881908 0.109460 -8.057 6.04e-15 ***
## I(lstat^2) 0.024918 0.003128 7.967 1.15e-14 ***
## I(rm^2) 0.973805 0.216680 4.494 8.73e-06 ***
## rad:lstat -0.035154 0.003714 -9.466 < 2e-16 ***
## rm:rad -0.338581 0.037780 -8.962 < 2e-16 ***
## crim:chas 2.141012 0.322928 6.630 8.90e-11 ***
## chas:nox -26.028896 5.684180 -4.579 5.93e-06 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.41 on 489 degrees of freedom
## Multiple R-squared: 0.8669, Adjusted R-squared: 0.8625
## F-statistic: 199 on 16 and 489 DF, p-value: < 2.2e-16
```

```
source("anova_alt.R")
anova_alt(fit.c)
```

```
## Analysis of Variance Table
##
##          Df    SS      MS      F        P
## Source  16 37030 2314.36 199.01 3.2656e-202
## Error   489  5687   11.63
## Total   505 42716   84.59
```

```
par(mfrow=c(2,2))
plot(fit.c)
```



```
par(mfrow=c(1,1))
```

Comment on the model fitted.

Answer:

Question 8

For the model fitted in Question 7, is the normality assumption reasonably satisfied?

Answer:

Question 9

Use Studentized Deleted Residual method to identify whether there are any outliers.

```
studresid = rstudent(fit.c)
studresid_sorted = sort(studresid)
head(round(studresid_sorted, 3), 10)
```

```
## 365 357 366 89 506 427 343 309 11 398
## -5.863 -3.372 -2.775 -2.530 -2.437 -2.129 -2.104 -1.887 -1.814 -1.809
```

```
tail(round(studresid_sorted,3),10)
```

```
##    370    153    392    187    162    182    373    410    369    372
## 2.261 2.365 2.481 2.567 2.802 3.365 3.583 4.022 4.281 7.780
```

Answer:

Question 10

Use Cook's Distance method to identify whether there are any influential observations.

```
cookdist = cooks.distance(fit.c)
cooks_sorted = sort(round(cookdist,3))

tail(round(cooks_sorted,3),10)
```

```
##    413    155    370    372    153    366    369    373    357    365
## 0.032 0.034 0.053 0.107 0.119 0.153 0.158 0.307 0.320 1.014
```

Answer: